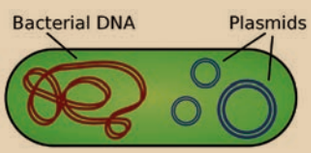
20.5 Genetic Engineering

Genetic engineering is the manipulation and transfer of genes from one organism to another organisms to create a new DNA called as **recombinant DNA(rDNA)**. The term recombinant is used because DNA from two different sources can be joined together. Hence, genetic engineering is also called as **recombinant DNA technology.**

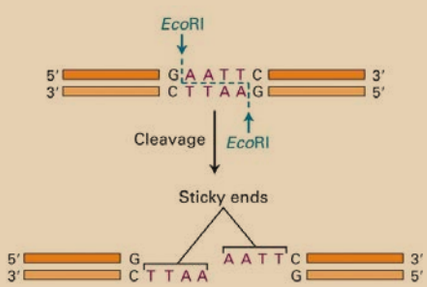
DO YOU KNOW?

Plasmid is the small circular double stranded DNA molecule found in the cytoplasm of bacterial cell and separated from chromosomal DNA. It can replicate independently.



More to Know

Restriction enzymes recognises a specific base pair sequence (palindromic sequence) in DNA called as restriction site and cleaves the phosphodiester bond within DNA.



20.5.1 Techniques of Genetic Engineering – Basic Requirements

Important discoveries that led to the stepping stone of rDNA technology were

a. Presence of **plasmid** in bacteria that can undergo replication independently along with chromosomal DNA.
b. **Restriction enzymes** cuts or break DNA at specific sites and are also called as molecular scissors.
c. **DNA ligases** are the enzymes which help in ligating (joining) the broken DNA fragments.

20.5.2 Gene Cloning

What reminds to your mind when you hear the word clone? Of course, 'DOLLY' the cloned sheep. The carbon copy of an individual is often called a **clone**. However, more appropriately, a clone means to make a **genetically exact copy of an organism**.

In gene cloning, a gene or a piece of DNA fragment is inserted into a bacterial cell where DNA will be multiplied (copied) as the cell divides. A brief outline of the basic steps involved in gene cloning are:

i. Isolation of desired DNA fragment by using restriction enzymes
ii. Insertion of the DNA fragment into a suitable vector (Plasmid) to make rDNA
iii. Transfer of rDNA into bacterial host cell (Transformation)
iv. Selection and multiplication of recombinant host cell to get a clone
v. Expression of cloned gene in host cell.

Using this strategy several enzymes, hormones and vaccines can be produced

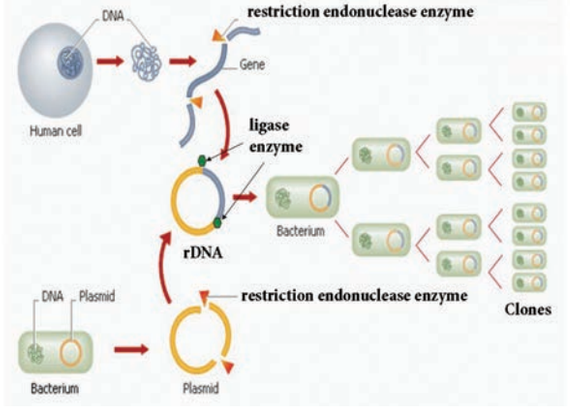


Figure 20.7 Genetic engineering technique (Gene cloning)

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